Analyzing Toronto Apartment Rental Prices

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Project Task Assignments

Raina wrote most of the introduction and data description.

Olivia did most of the exploratory data analysis section.

Austin (group leader) did most of the modeling section.

Woobin wrote most of the conclusion.

All members contributed to editing and proofreading the project as a whole.

Introduction

In high-density metropolitan areas like Toronto, assessing how specific layout configurations influence rental pricing provides valuable insight into market structure and housing affordability. This study examines the relationship between apartment layout characteristics and monthly rent across the city. Specifically, we use multiple linear regression to evaluate how the number of bedrooms, bathrooms, and their interaction effects predict continuous rental prices. To test predictive validity further, we apply binary logistic regression using the same structural features to determine whether a listing classifies as a high-value property. By comparing main-effects and interaction models across both regression frameworks, we assess the overall explanatory power of layout features in Toronto's housing market.

Data Description

The dataset for this study was retrieved from a public Kaggle repository titled Toronto Apartment Rental Price, compiled by independent creator Raja CSP. According to the dataset documentation, the listings were aggregated and scraped from local rental websites active across Toronto in 2018.

The study utilizes Version 3 of the dataset, stored as a single 94.97 KB comma-separated values file. The raw file contains `n = 1,124` observational rows across 7 attribute columns, where each row corresponds to a unique apartment listing. There are no missing values in any columns in the raw data.

Exploratory Data Analysis

During data cleaning, we removed 308 duplicates which likely occurred because the data came from multiple sources. We also removed 9 rows where `price` was either below \$100/month or above \$6,000/month (namely, \$65, \$99, \$99, \$6,000, \$6,500, \$8,000, \$9,750, \$36,900, and \$535,000) because these outliers may skew our results. The sample size is now `n = 807`.

Before modeling, we want to find out more about the data.

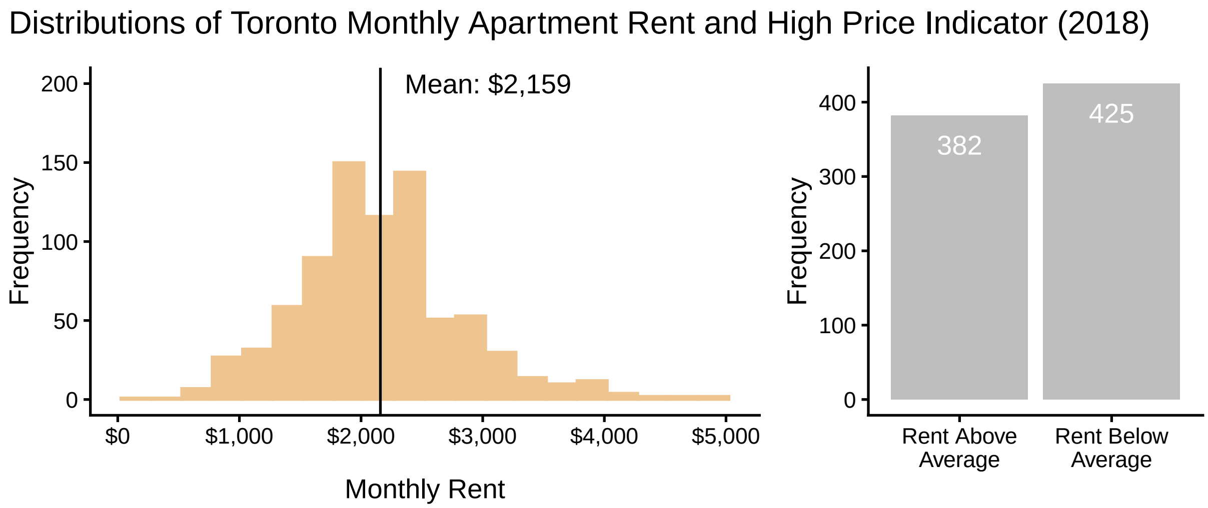
We first examine sample's summary statistics of each individual numerical variable.

Variable	Mean	Standard Deviation	Minimum	Maximum
bathroom	1.23	0.43	1.00	3.00
bedroom	1.36	0.56	1.00	3.00
den	0.15	0.36	0.00	1.00
lat	43.71	0.69	42.99	56.13
long	-79.50	1.85	-114.08	-73.58
price	2159.05	684.27	150.00	4900.00

The amount of `bathroom`s ranges from 1 to 3, the amount of `bedroom`s ranges from 1 to 3, and the amount of `den`s ranges from 0 to 1. The average amount of bedrooms is slightly higher than the average amount of bathrooms, and most apartment's don't have any dens, shown by its mean of 0.15. This makes sense - an apartment should have a bedroom and bathroom whereas a den is optional. Also, `latitude` and `longitude` fall in the valid range of [-90, 90] and [-180, 180] respectively (though we won't use these variables in modeling). So there's nothing unusual about these four variables.

As for `price`, its average value is \$2,159/month with a standard deviation of \$684/month. Its minimum is \$150/month and its maximum is \$4,900/month. While these minimum and maximum values are -2.59 and 4.01 standard deviations above and below the mean (respectively), we feel that they are reasonable enough to keep.

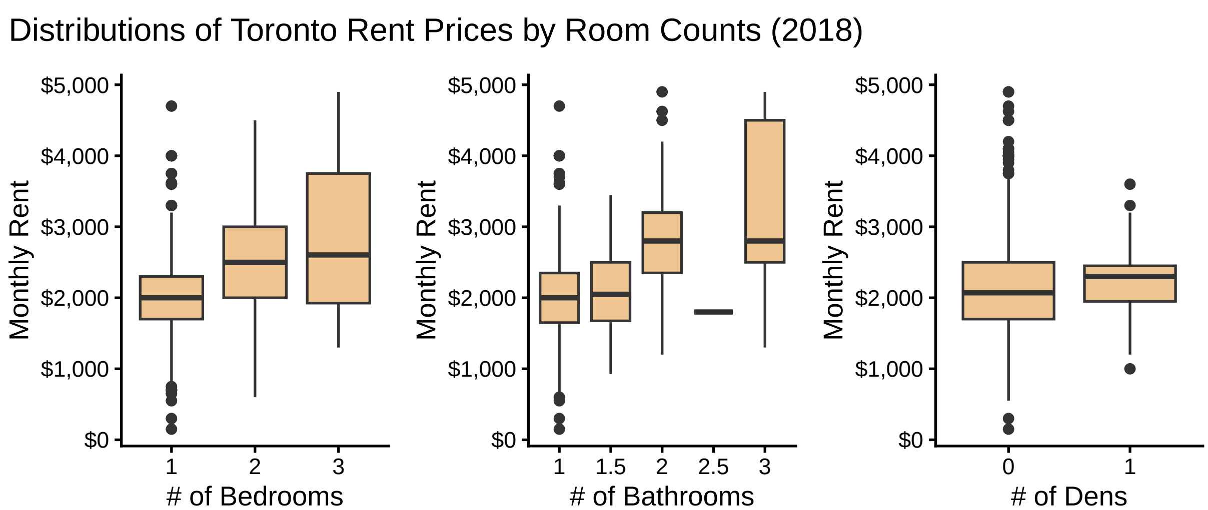
We now plot a histogram to visualize the distribution of `price` and a bar chart showing the class balance of the new binary variable `high_price`.



On the left, we see that `price` is approximately symmetric and unimodal around its mean. On the right, we see that the two `high_price` groups are approximately balanced; this means that logistic regression won't run into the issue of unbalanced response groups, which could bias modeling results.

Next, we examine possible relationships *between* variables.

Below are boxplots of `price` split by `bedroom`, `bathroom`, and `den` counts to see how `price`'s distribution varies by each of these apartment characteristics.



Looking independently at `bedroom`, `price`'s distribution slightly shifts upwards by a few hundred dollars for each additional bedroom. For `bathroom`, the distribution also moves upwards with the biggest drop from 2 to 2.5 bathrooms and another upwards shift in the distribution from 2.5 to 3 bathrooms. For `den`, the distribution is roughly equal for both groups with a difference-between-means of only $87/month. As well, within each group of boxplots, the distributions overlap with each other, so these predictors may not produce strong modeling results. Nonetheless, in terms of potential modeling power, `bathroom` may be better than `bedroom` and it may be better not to use `den`.

We're also curious about the correlations between the variables so we plot the upper triangle of a (Pearson) correlation matrix. Pearson correlation, denoted by the letter r, falls in [0, 1]. If two variables are positively correlated, it means that as one gets larger, the other also tends to get larger; if two variables are negatively correlated, it means that as one gets larger, the other tends to get smaller.

Looking at the rightmost column, the variables with the strongest linear correlation with `price` are `bathroom` (r = 0.47) then `bedroom` (r = 0.39), though these are only moderately strong. In addition to the boxplots' evidence, the matrix further confirms that `den` is weakly correlated with `price` at r = 0.05. And even though `bedroom` and `bathroom` are moderately correlated at r = 0.63, putting models at risk of multicollinearity, both variables measure different characteristics of an apartment, so we may keep both during modeling.

In addition, here are frequency tables between `high_price` and `bedroom` and `bathroom`, excluding `den` because we've already seen that it would be a weak predictor.

# of Bedrooms	Rent Below Average	Rent Above Average	Total
1	62% (340)	38% (208)	100% (548)
2	32% (72)	68% (153)	100% (225)
3	38% (13)	62% (21)	100% (34)
Total	53% (425)	47% (382)	100% (807)

# of Bathrooms	Rent Below Average	Rent Above Average	Total
1	62% (381)	38% (232)	100% (613)
1.5	52% (12)	48% (11)	100% (23)
2	18% (30)	82% (135)	100% (165)
2.5	100% (1)	0% (0)	100% (1)
3	20% (1)	80% (4)	100% (5)
Total	53% (425)	47% (382)	100% (807)

Most of the data is on 1-bedroom or 1-bathroom units with the least data on units with 1.5, 2.5, or 3 bathrooms. Expectedly, most (62%) of the below-average units have 1 bedroom and 1 bathroom.

The main findings of this exploratory analysis are:

1. There is no class imbalance in `high_price` (for the purpose of logistic regression) because 53% vs. 47% of the observations have a `price` above vs. below average. Class imbalance, such as 90% vs. 5%, could bias a model's intercept and produce a poor model that maximizes accuracy by simply predicting the majority class.
2. In terms of predictive ability, `den` likely has the weakest linear relationship with `price`, as shown by the very similar boxplots and Pearson correlation of 0.05, and we exclude `latitude` and `longitude` because geographic effects on a target variable are probably not as linear or straightforward as these coordinates. This leaves `bedroom` and `bathroom` as the covariates remaining.

Models

Now that we have explored the data, we can consider models that can model rental price based on other variables. In the previous section, we narrowed down the variables we can use as predictors to the number of bedrooms and the number of bathrooms. To make use of these predictors, we chose two model types: a multiple linear regression on `price` because it is continuous and a multiple logistic regression on `high_price` because it is binary.

Multiple Linear Regression

A multiple linear regression model models a continuous response variable in terms of the predictors as a straight line with an intercept and at least one slope.

When making this model type, we often consider whether interaction between covariates is statistically significant. If it is, then the slope of the prediction line differs depending on the value of the covariates, otherwise the slope is constant. Statistical significance can be checked with hypothesis testing by creating a model with interaction and checking its statistics.

Results with interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1496.02	210.86	7.09	0.00	1082.12	1909.93
bedroom	0.95	125.45	0.01	0.99	-245.30	247.21
bathroom	318.44	180.91	1.76	0.08	-36.68	673.56
bedroom:bathroom	146.84	90.94	1.61	0.11	-31.67	325.35

Results without interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1172.76	66.25	17.70	0	1042.72	1302.80
bedroom	187.90	48.35	3.89	0	93.00	282.80
bathroom	592.21	63.16	9.38	0	468.23	716.19

In first table, the interaction term (`bedroom:bathroom`) has a p-value of 0.11 in excess of our pre-decided level of significance of 0.05. Thus, we cannot confidently identify any interaction between the predictors and choose to proceed with the model without interaction.

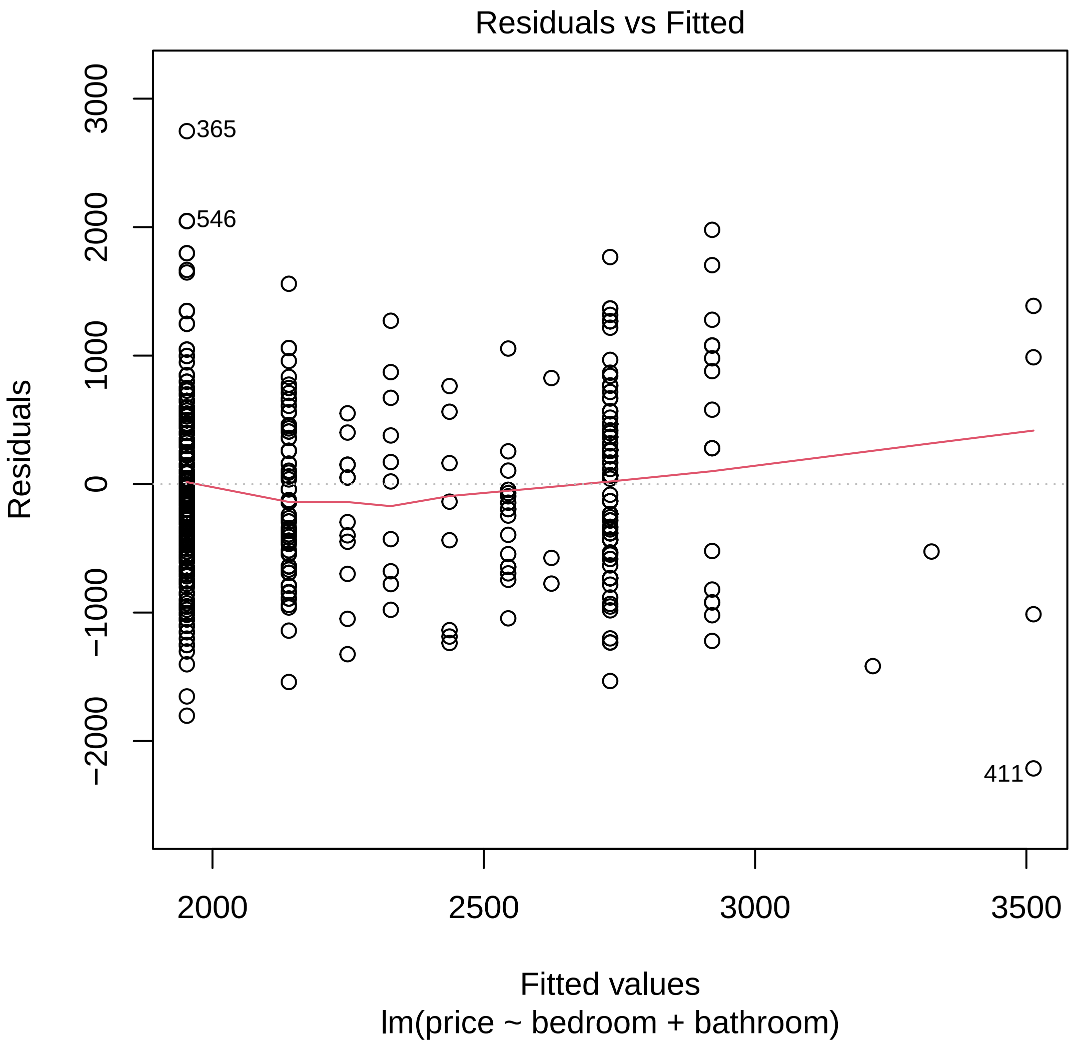
The `estimate` column in the table for the non-interaction model shows a prediction line of the form:

$$\text{price} = \$1172.76 + \$187.90 \cdot \text{bedroom} + \$592.21 \cdot \text{bathroom}$$

Its intercept shows that the baseline prediction for a unit with no bathrooms or bedrooms is \$1172.76, which could represent costs for utilities and other living areas, like the kitchen. The slope values show that an additional bathroom in a property is associated with a \$187.90 increase in monthly rent when the number of bathrooms is held constant at any value, and that an additional bathroom is associated with a \$592.21 increase in rent when the number of bedrooms is held constant at any value. For example, the model predicts that a one-bed, one-bath condo has a rental price of \$1172.76 + \$187.90 + \$592.21 = \$1952.87.

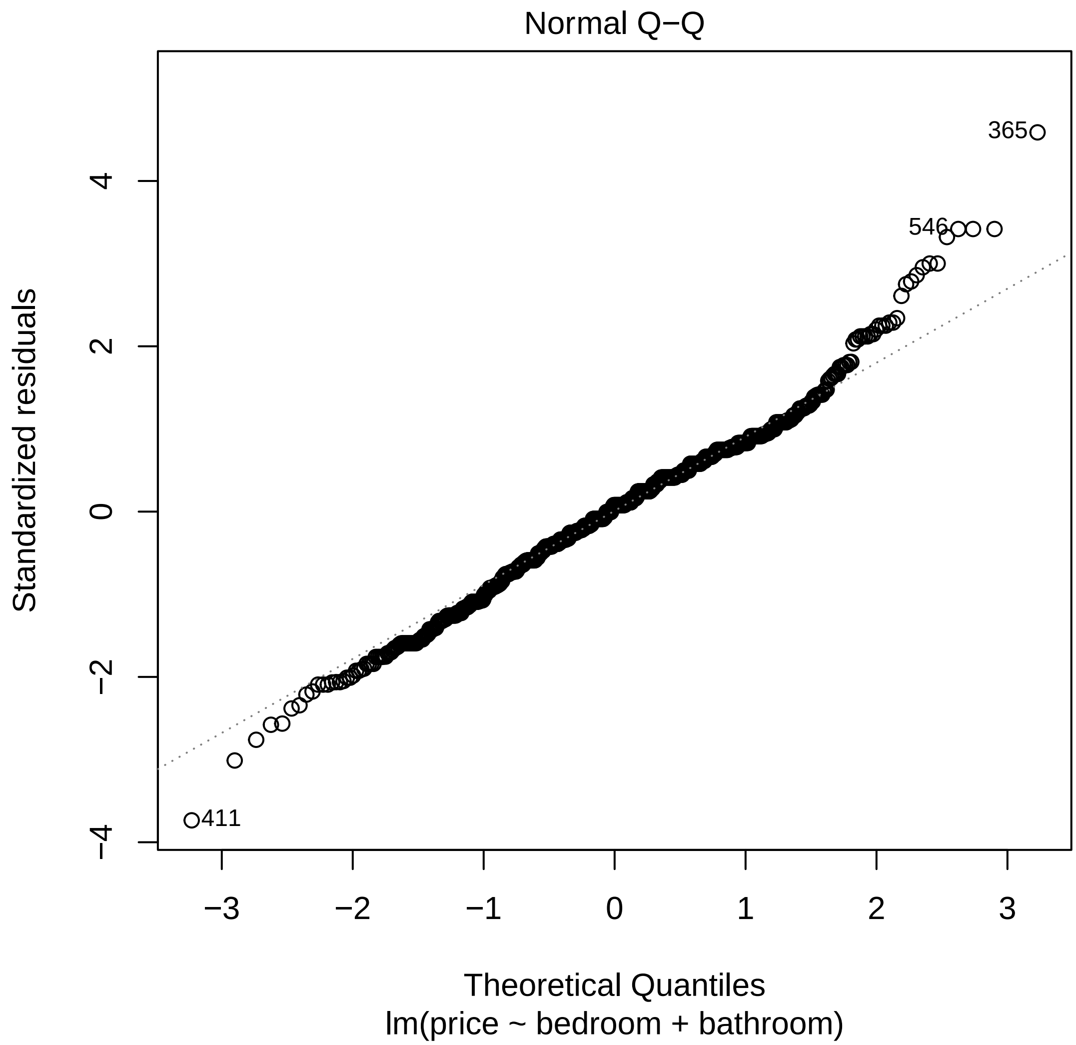
To check if the fitted model is appropriate, we run model diagnostics in the form of a residual plot and a Q-Q plot.

TODO: explain what a residual plot is



For the fitted model to be appropriate, the error around the trend line should be consistent and normally distributed. Residual plots visualize these errors. In a residual plot, errors should be distributed normally and evenly around a horizontal line at r = 0. Because our predictors are discrete rather than continuous, the data points are not continuously distributed, so the plot is a bit messy. While the error approximately follows a normal distribution, there is larger variance around certain areas in the x-axis, so the variance is not perfectly consistent.

Another way of checking the distribution of the error is via a Q-Q plot. If the error is normally distributed with equal variance, then a Q-Q plot will show all data points distributed along a straight, sloped line wherein x = y.



The Q-Q plot of our linear model shows the data following this line for the most part, but diverging at the beginning and the end, implying that our model has more extreme data than can be expected under a true normal distribution (Clay, 2015). Applying transformations to the model, such as log transformation and predicting the square root of the response variable, were unsuccessful in creating a more favourable distribution.

While our diagnostics reveal that our data may not be perfectly suited for a linear analysis, this is not uncommon for observed data. It is our belief that our model can still be useful for predicting.

Multiple Logistic Regression

A multiple logistic regression model is used to make predictions for a binary response variable. To make use of such a model, we converted our response variable (price) into a binary one by splitting it into two categories for rent above and below the mean value. Thus, our model will be predicting whether a property has above or below average rent, rather than a specific value.

Once again, we need to begin by checking for interaction between predictors.

Results with interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.30	0.17	-1.82	0.07	-0.62	0.02
bedroom	0.20	0.10	2.05	0.04	0.01	0.39
bathroom	0.58	0.14	4.13	0.00	0.31	0.86
bedroom:bathroom	-0.12	0.07	-1.70	0.09	-0.26	0.02

Results without interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.04	0.05	-0.68	0.50	-0.14	0.07
bedroom	0.05	0.04	1.26	0.21	-0.03	0.12
bathroom	0.36	0.05	7.27	0.00	0.26	0.46

Our results show that, much like for our linear model, we cannot conclude that there is significant interaction between our predictors. For that reason, we will proceed with our model without interaction.

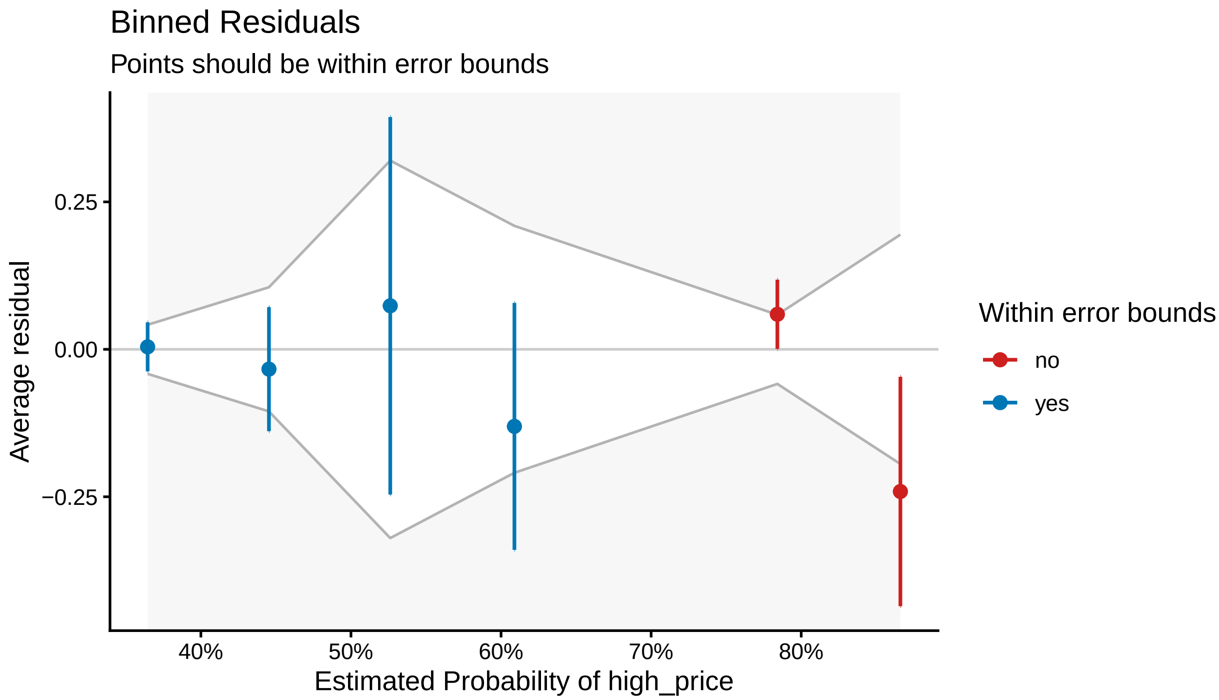
The table of our model without interaction shows that an additional bedroom increases the odds of a property having high rent by 1.24 or 24%. However, the p-value for this estimate implies that this conclusion is not statistically significant. The model also found that an additional bathroom increases the odds of a property having high rent by 5.25 or 425%, a finding that is statistically significant.

Again we must run some model diagnostics to ensure our model is appropriate for the data.

One of the most important things to check for is overdispersion. Our models assume that the distribution of our models response is consistent with a Bernoulli distribution. Overdispersion occurs when our response distribution deviates from this assumption. Our model has a dispersion parameter of 1.025, implying insignificant levels of overdispersion.

TODO: maybe precision and accuracy metrics

As an additional assessment, we use a binned residuals plot. In a binned residuals plot, the majority of data points (approximately 95%) must fall within the generated error bounds for the model to be considered appropriate for the data (Ludecke et al., n.d.).



As you can see, our model easily meets this criteria as all of the data points fall within these bounds.

Conclusion

The multiple linear regression analysis indicates that both the number of bedrooms and bathrooms have a positive relationship with monthly rent. Specifically, holding other variables constant, an additional bedroom increases predicted monthly rent by \$187.90 ($p<0.001$), while an additional bathroom increases rent by \$592.21 ($p<0.001$). The multiple logistic regression analysis indicates that bathrooms have a statistically significant positive effect on the odds of a property having an above-average rent ($p<0.001$), increasing the odds by 43% per bathroom, while the effect of bedrooms is not statistically significant at the standard 0.05 level ($p=0.21$)

However, as seen from the residual plot and the Q-Q plot, the key assumptions of homoscedasticity and the normality of residuals for the linear model are invalid. These violations affect prediction confidence intervals and the p-values. Namely, the current MLR model results cannot be considered fully reliable. Consequently, findings regarding property characteristics should be interpreted with caution given potential omitted variable bias.

To improve model reliability in future research, three key adjustments are recommended: Re-introducing location through neighbourhood clusters will help reduce unexplained variance and standard errors, thereby addressing potential omitted variable bias. Collecting continuous structural variables like square footage alongside discrete room counts will help normalize the residual distribution. Standard 95% confidence thresholds should be maintained to ensure hypothesis testing remains statistically unbiased.

References

References will go here.

Appendix

```r
# Loading packages
library(tidyverse)
library(knitr)
library(scales)
library(patchwork)
library(corrplot)
library(broom)
```